

ECE 105: Introduction to Electrical Engineering

Lecture 10

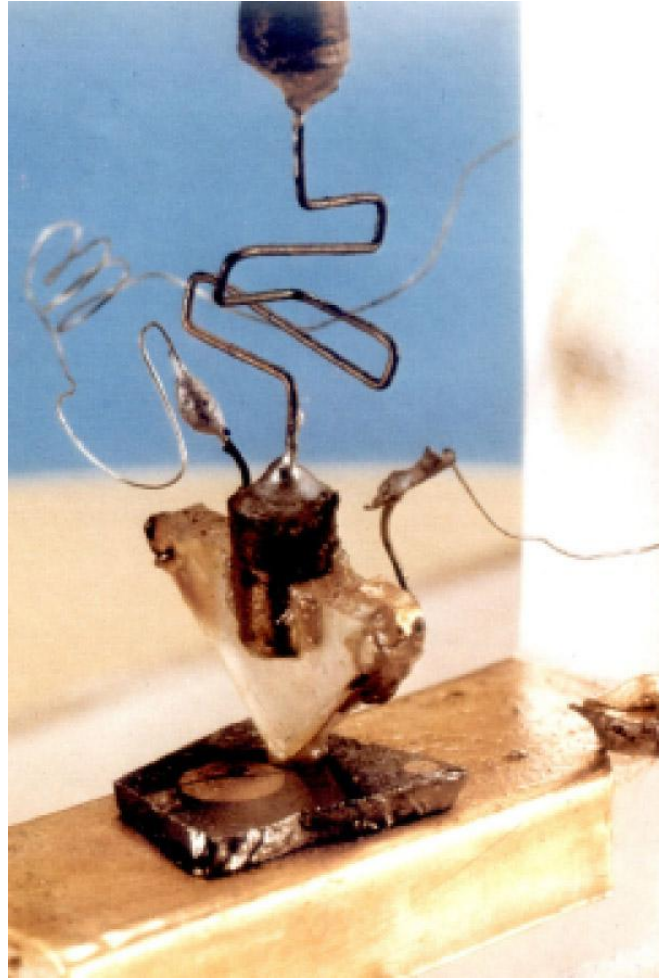
Device 2

Yasser Khan

Rehan Kapadia

The First Transistor

- At **Bell Telephone Laboratories**, research was focused on finding a solid-state amplifier for telephone signal processing.
- The three key scientists were:
 - John Bardeen** (theorist)
 - Walter Brattain** (experimentalist)
 - William Shockley** (team leader, physicist)



- December 16, 1947**: John Bardeen and Walter Brattain successfully demonstrated the **first working transistor** at Bell Labs.
- It was a **point-contact transistor** made of a small slab of germanium with two closely spaced gold contacts.
- When input current was applied to one contact, it modulated the conductivity of the germanium, amplifying the signal seen at the other contact.
- Key outcome**: It could amplify electrical signals like a vacuum tube, but in a device only a few millimeters wide.

Transistor amplification

1956 Physics Nobel Prize

"for their researches on semiconductors and their discovery of the transistor effect"



William Bradford Shockley

🕒 1/3 of the prize

USA

Semiconductor
Laboratory of Beckman
Instruments, Inc.
Mountain View, CA, USA



John Bardeen

🕒 1/3 of the prize

USA

University of Illinois
Urbana, IL, USA



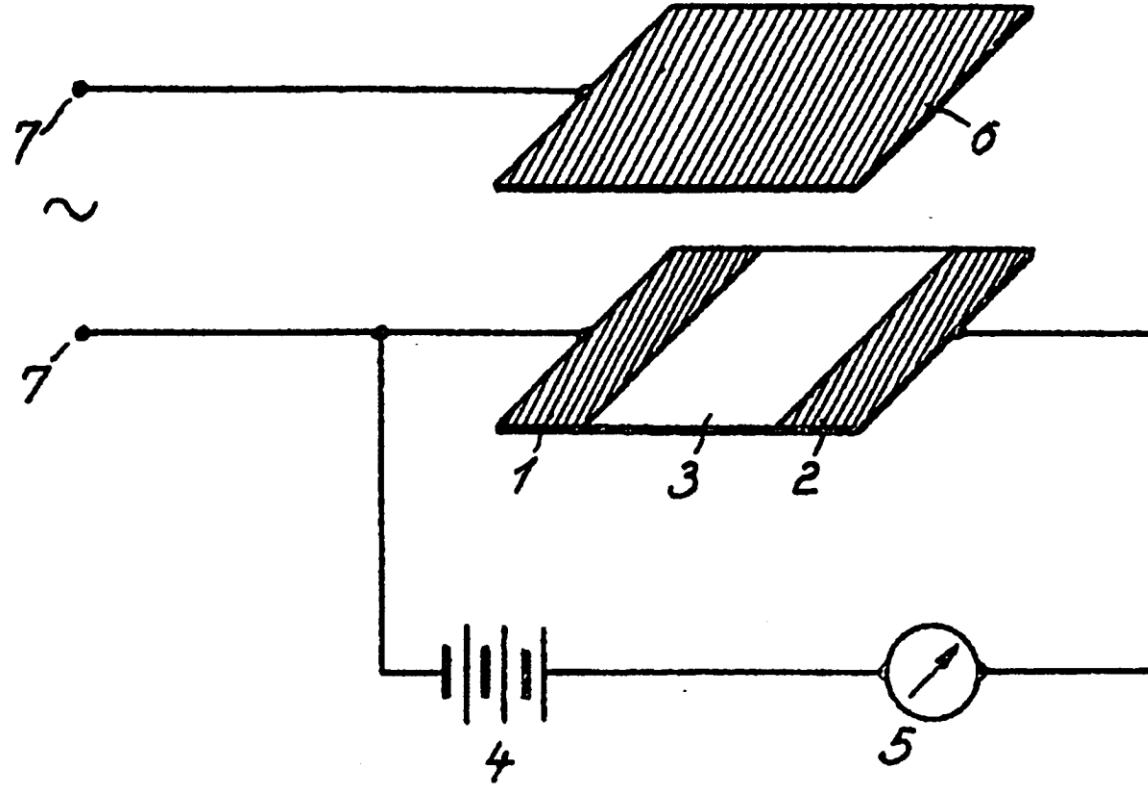
Walter Houser Brattain

🕒 1/3 of the prize

USA

Bell Telephone
Laboratories
Murray Hill, NJ, USA

Invention of the Field-Effect Transistor

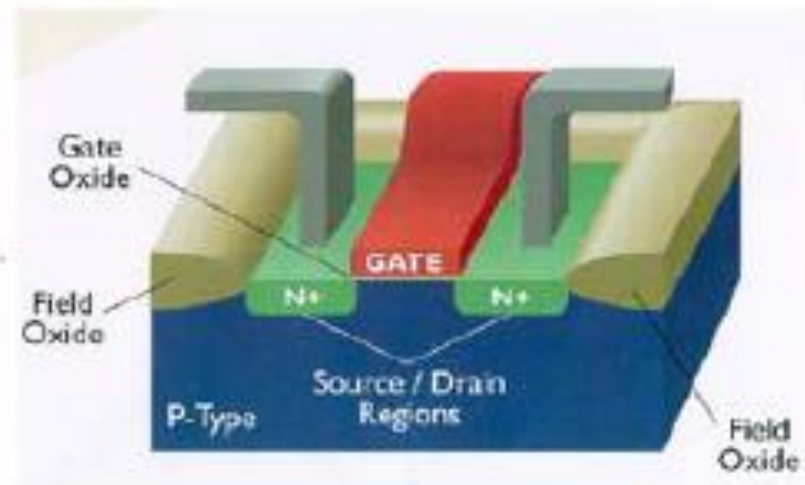


In 1935, a British patent was issued to Oskar Heil.
A working MOSFET was not demonstrated until 1955.

Modern Field Effect Transistor (FET)

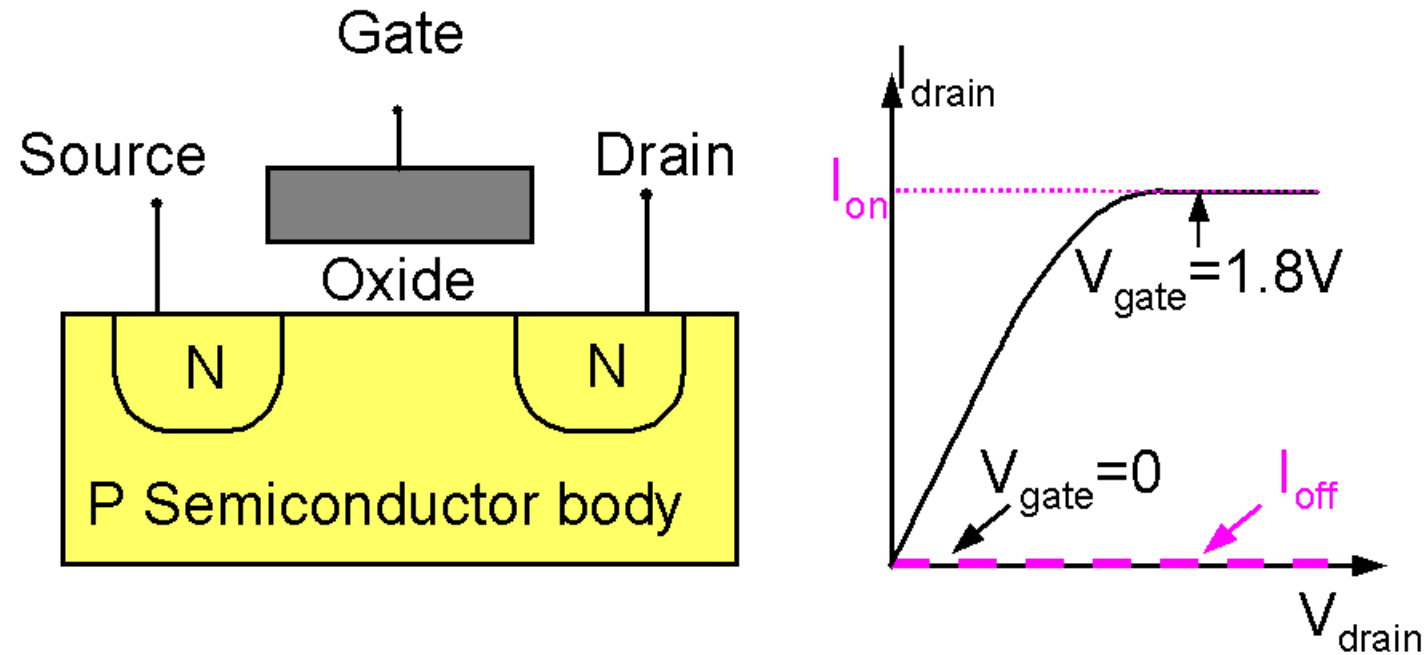
- An electric field is applied normal to the surface of the semiconductor (by applying a voltage to an overlying electrode), to modulate the conductance of the semiconductor
- Modulate drift current flowing between 2 contacts (“source” and “drain”) by varying the voltage on the “gate” electrode

N-channel MOSFET:



Introduction to the MOSFET

Basic MOSFET structure and IV characteristics

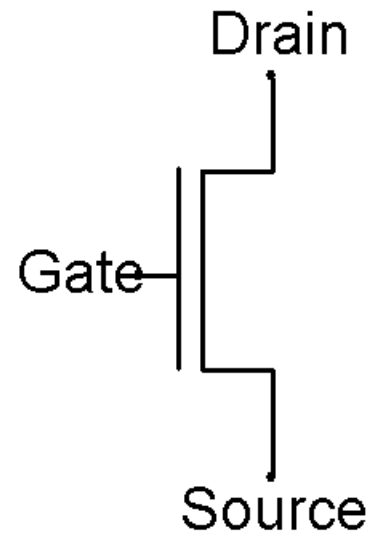


What is desirable: large I_{on} , small I_{off}

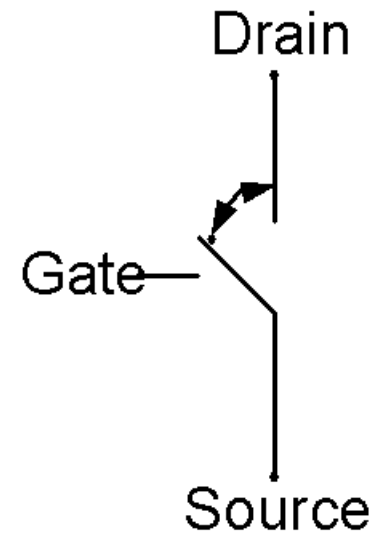
Introduction to the MOSFET

Two ways of representing a MOSFET:

Circuit Symbol

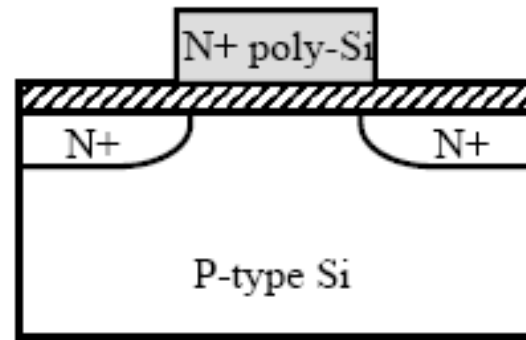


Simple Switch



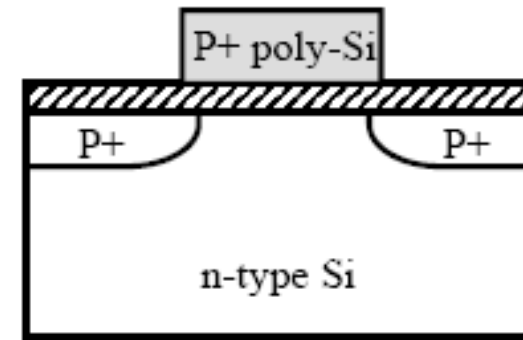
N-channel vs. P-channel

NMOS



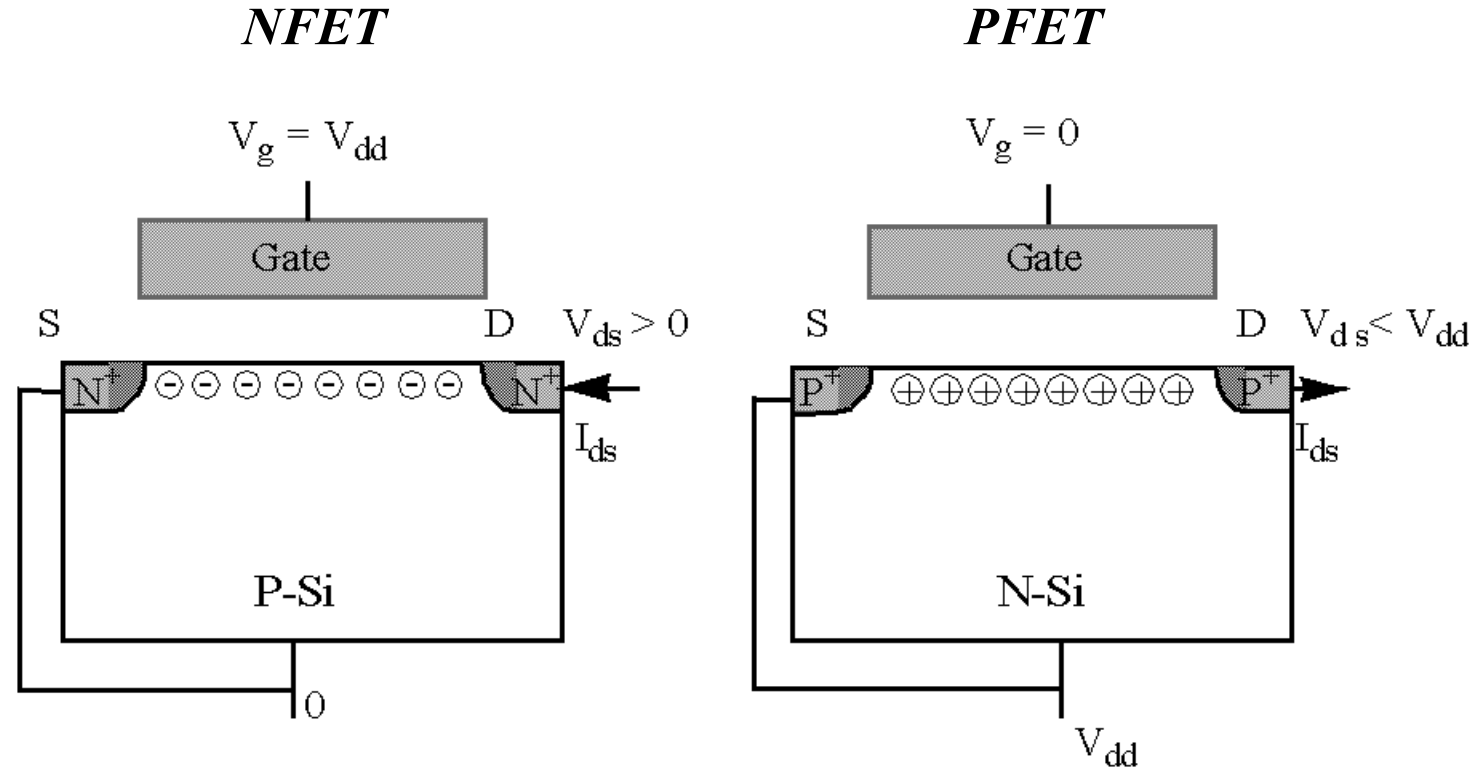
- For current to flow, $V_{GS} > V_T$
- Enhancement mode: $V_T > 0$

PMOS



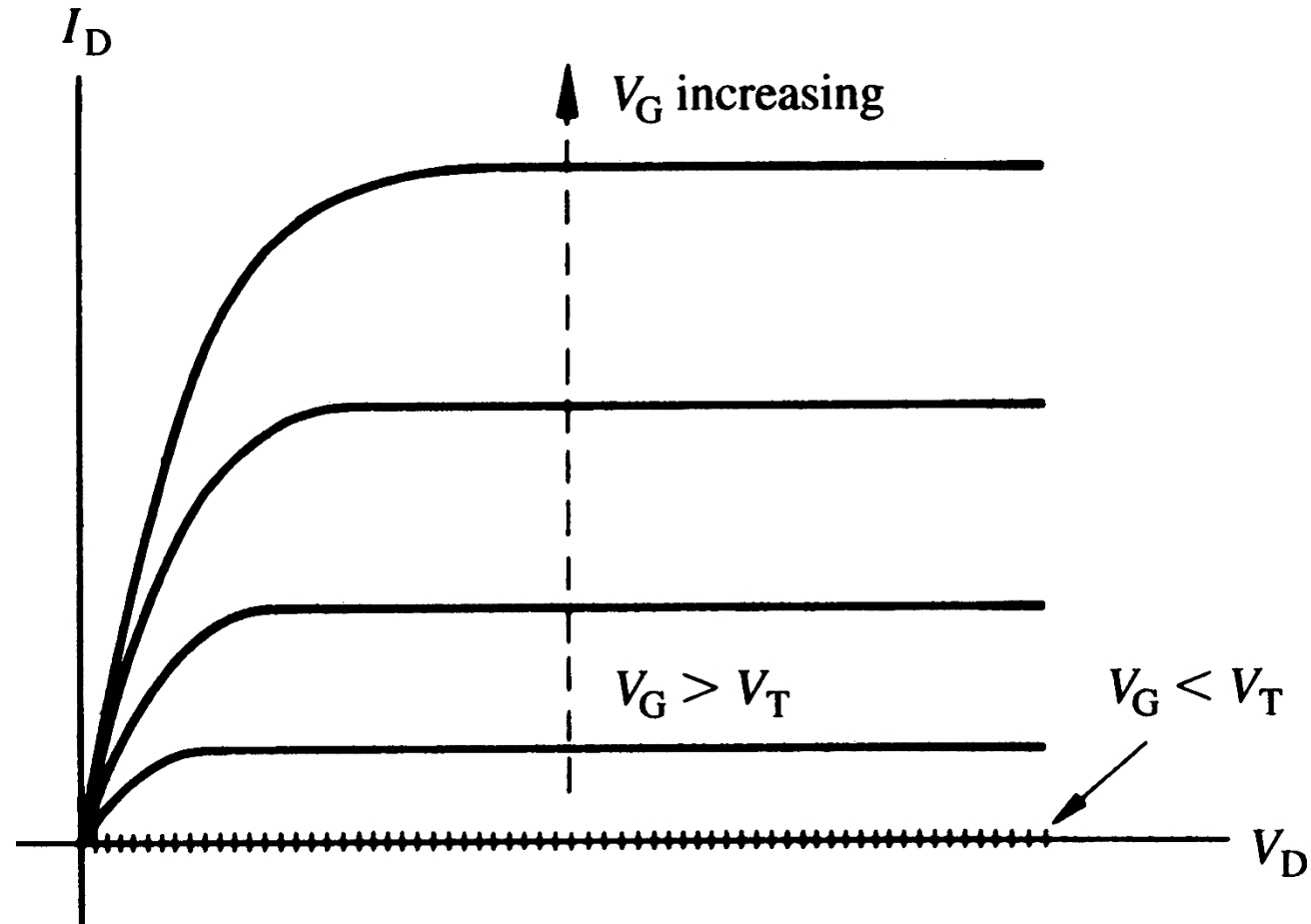
- For current to flow, $V_{GS} < V_T$
- Enhancement mode: $V_T < 0$

Complementary MOSFETs (CMOS)



*When $V_g = V_{dd}$, the N-FET is on and the P-FET is off.
When $V_g = 0$, the P-FET is on and the N-FET is off.*

I_D - V_{DS} characteristics expected from a long channel
($\Delta L \ll L$) MOSFET (n-channel), for various values of V_G



MOSFET current equations

MOSFET Equations

- Linear (ohmic) region:

$$I_D = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

(valid when $V_{DS} < V_{GS} - V_{TH}$)

- Saturation region:

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

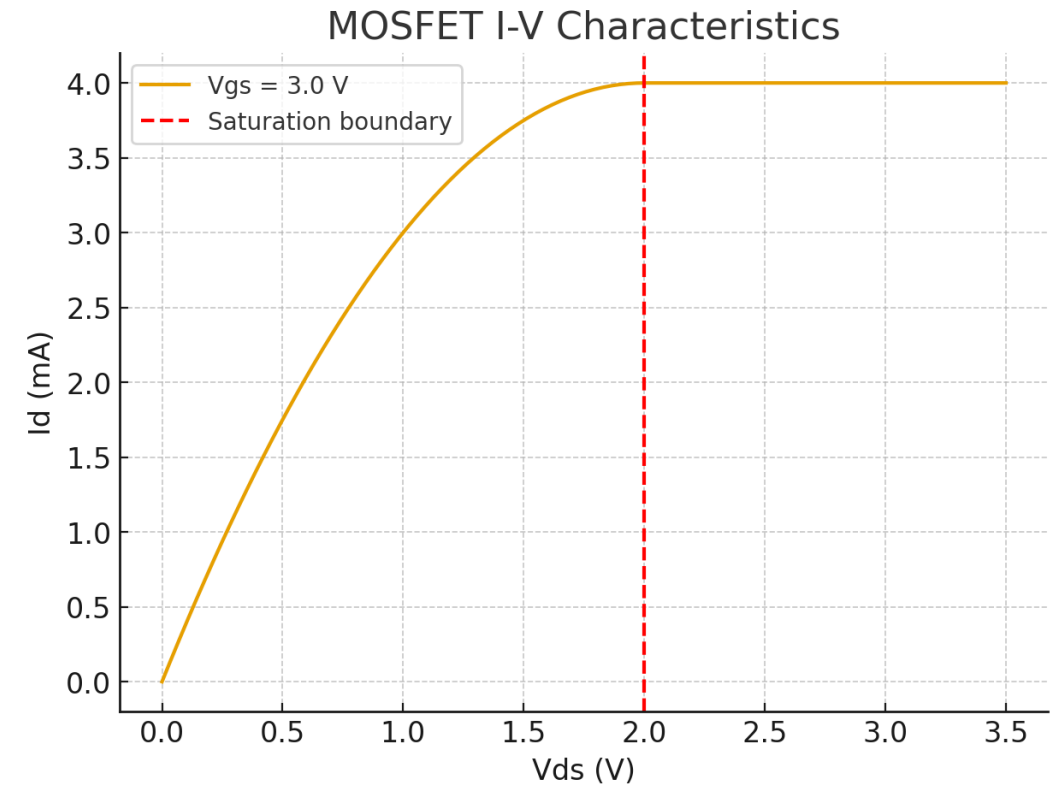
(valid when $V_{DS} \geq V_{GS} - V_{TH}$)

Assume Device Parameters

- Mobility-oxide product: $\mu C_{ox} = 200 \mu A/V^2$
- Width/Length ratio: $W/L = 10$
- Threshold voltage: $V_{TH} = 1.0 V$

So,

$$k' = \mu C_{ox} \frac{W}{L} = 200 \times 10 = 2000 \mu A/V^2 = 2 mA/V^2$$



MOSFET current equations

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Example 1 — Linear region

- $V_{GS} = 3.0 V$
- $V_{DS} = 0.5 V$

Check region: $V_{GS} - V_{TH} = 2.0 V$. Since $V_{DS} = 0.5 < 2.0$, device is in linear region.

$$\begin{aligned} I_D &= 2 \left[(2.0)(0.5) - \frac{1}{2} (0.5)^2 \right] \\ &= 2 [1.0 - 0.125] = 2 \times 0.875 = 1.75 mA \end{aligned}$$

MOSFET current equations

MOSFET Equations

- Linear (ohmic) region:

$$I_D = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

(valid when $V_{DS} < V_{GS} - V_{TH}$)

- Saturation region:

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

(valid when $V_{DS} \geq V_{GS} - V_{TH}$)

Example 2 — Saturation region

- $V_{GS} = 3.0 \text{ V}$
- $V_{DS} = 3.0 \text{ V}$

Check region: $V_{GS} - V_{TH} = 2.0 \text{ V}$. Since $V_{DS} = 3.0 \geq 2.0$, device is in **saturation**.

$$\begin{aligned} I_D &= \frac{1}{2} \times 2 \times (2.0)^2 \\ &= 1 \times 4.0 = 4.0 \text{ mA} \end{aligned}$$

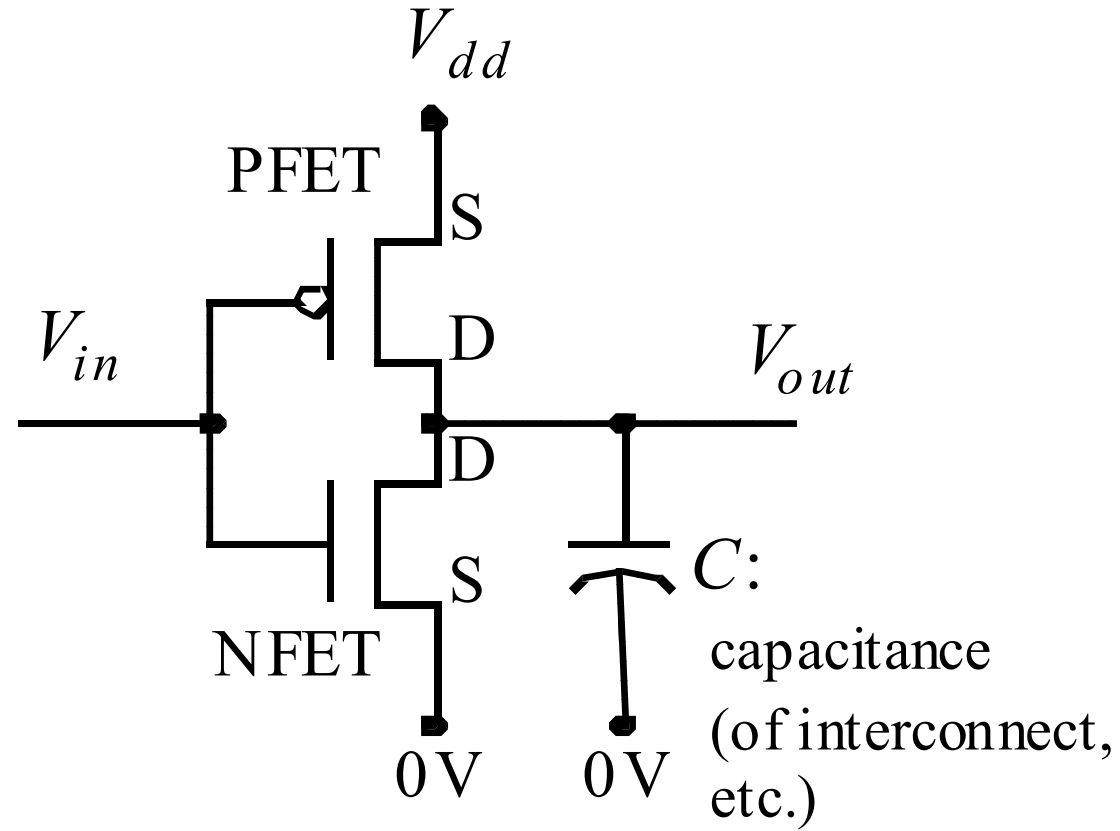
Assume Device Parameters

- Mobility–oxide product: $\mu C_{ox} = 200 \mu\text{A}/\text{V}^2$
- Width/Length ratio: $W/L = 10$
- Threshold voltage: $V_{TH} = 1.0 \text{ V}$

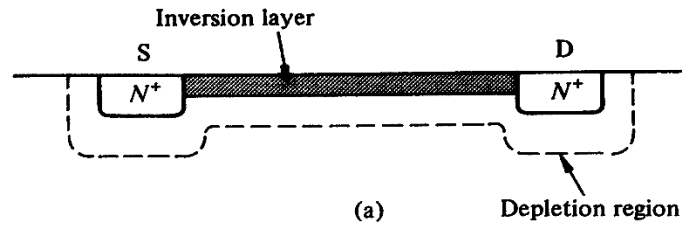
So,

$$k' = \mu C_{ox} \frac{W}{L} = 200 \times 10 = 2000 \mu\text{A}/\text{V}^2 = 2 \text{ mA}/\text{V}^2$$

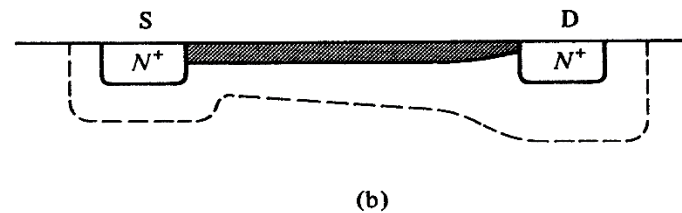
CMOS (Complementary MOS) Inverter



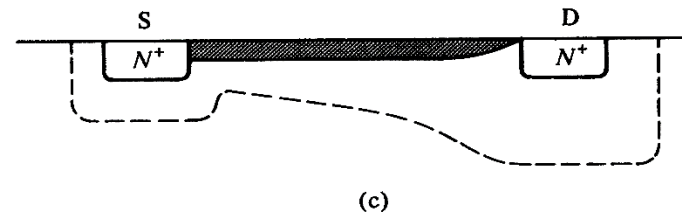
Qualitative discussion: *n*-MOSFET



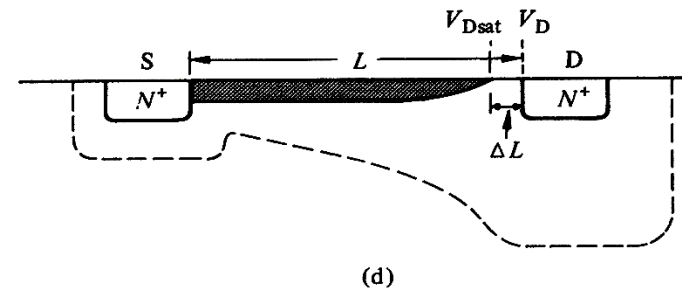
$V_G > V_T$; $V_{DS} \approx 0$
 I_D increases with V_{DS}



$V_G > V_T$; V_{DS} small, > 0
 I_D increases with V_{DS} , but
rate of increase decreases.



$V_G > V_T$; $V_{DS} \approx \text{pinch-off}$
 I_D reaches a saturation value, $I_{D,sat}$
The V_{DS} value is called $V_{DS,sat}$



$V_G > V_T$; $V_{DS} > V_{DS,sat}$
 I_D does not increase further,
saturation region.

Quantitative I_D - V_{DS} Relationships – 1st attempt

“Square Law”

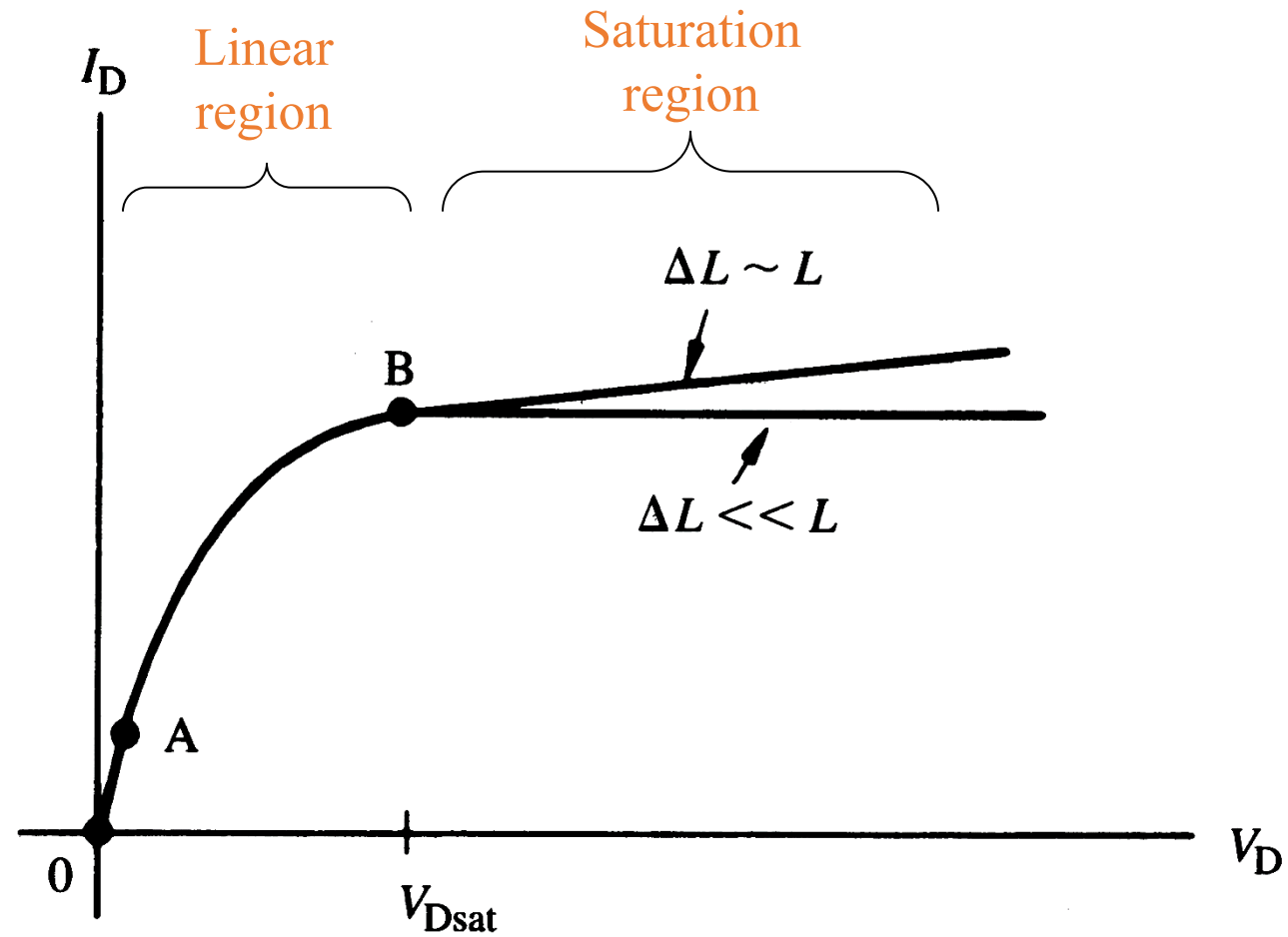
$$I_D = \mu C_{ox} \frac{W}{L} [(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2] \quad 0 < V_{DS} < V_{DS,sat} \quad ; \quad V_G > V_T$$

I_D will increase as V_{DS} is increased, but when $V_G - V_{DS} = V_T$, pinch-off occurs, and current saturates when V_{DS} is increased further. This value of V_{DS} is called $V_{DS,sat}$. i.e., $V_{DS,sat} = V_G - V_T$ and the current when $V_{DS} = V_{DS,sat}$ is called $I_{DS,sat}$.

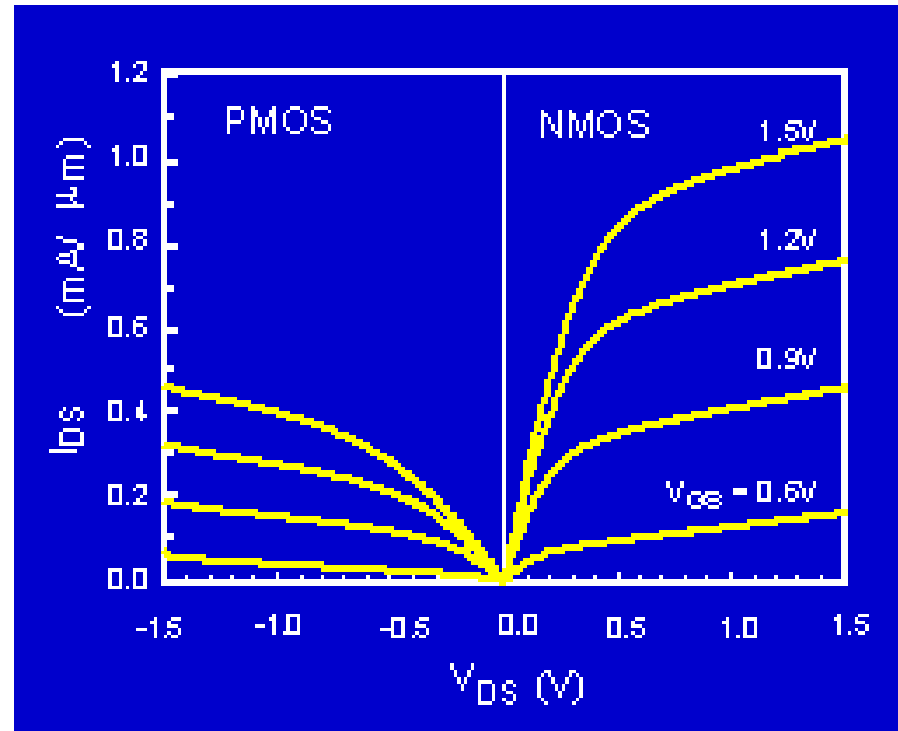
$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad V_D > V_{DS,sat} \quad ; \quad V_G > V_T$$

Here, C_{ox} is the oxide capacitance per unit area, $C_{ox} = \epsilon_{ox} / x_{ox}$

I_D - V_{DS} characteristics for n -MOSFET

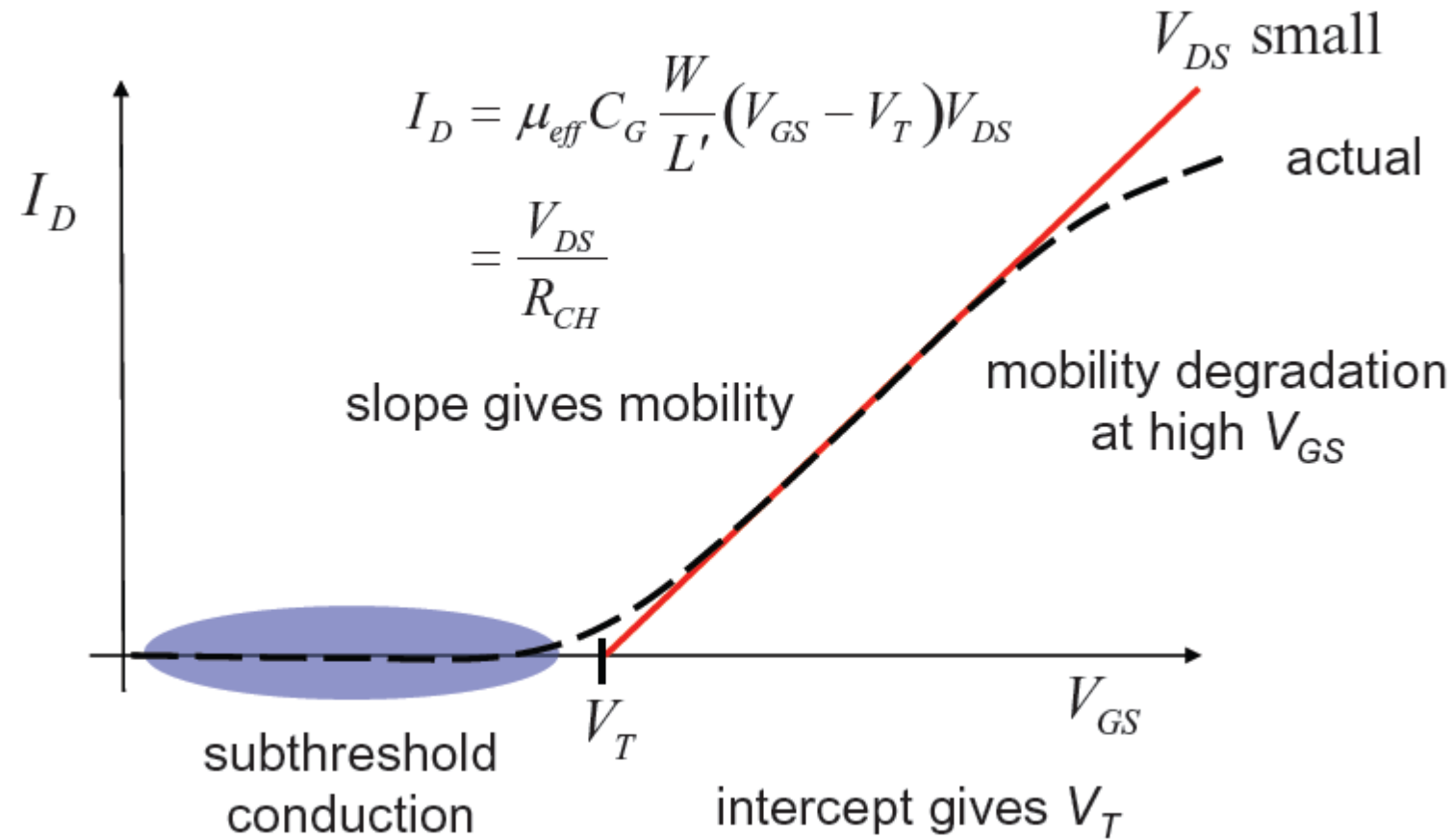


P-MOSFET N-MOSFET IV Characteristics

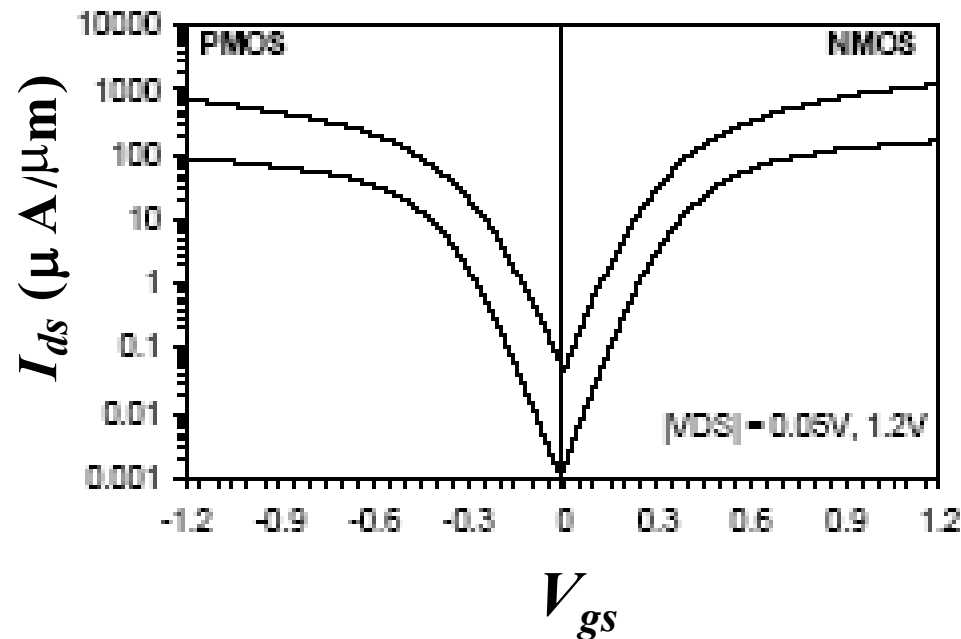


The PMOS IV is qualitatively similar to the NMOS IV, but the current is about half as large. Why?

Threshold and Subthreshold



- The leakage current that flows at $V_g < V_t$ is called the subthreshold current. Previously we had assumed that current is zero, but in reality that's not the case.



Intel, T. Ghani et al., IEDM 2003

90nm technology.
Gate length: 45nm for
NMOS, 50nm for
PMOS

- The current at $V_{gs}=0$ and $V_{ds}=V_{dd}$ is called I_{off} .